

Comments on <https://vidal-lab.ding.tj/>

I love the Project highlights. We should ask people to create one for each person or one per project so we can have them. These could go in a rotating set of pictures on the landing page.

Comments on <https://vidal-lab.github.io/>

We should add some Google Scholar page for the LAB

https://scholar.google.com/citations?hl=en&view_op=search_authors&mauthors=Hongkang_Li+OR+Lachlan_MacDonald+OR+Nghia_Nguyen+OR+kyle_poe+OR+Konstantinos_Emmanouilidis+OR+Tadipatri_Uday+OR+Buyun_Liang+OR+benjamin_haeffele+OR+yuyan_ge+OR+ziqing_xu+OR+fengrui_tian+OR+darshan_thaker+OR+jinqi_luo+OR+Kwan_Ho_Ryan_Chan+OR+tianjiao_ding+OR+rene_vidal+OR+salma_tarmoun+OR+liangzu_peng+OR+kaleab_kinfu+upenn&btnG=

HOME

VIDAL Lab

Vision, Intelligence, Dynamics and Learning Laboratory

Departments of ESE & Radiology · University of Pennsylvania

About Us

The VIDAL (Vision, Intelligence, Dynamics and Learning) Lab develops the mathematical and algorithmic foundations of modern artificial intelligence, with an emphasis on developing trustworthy AI systems with real-world impact. Our research philosophy is to combine mathematical foundations, scalable algorithms, and real-world deployment to build AI systems that are both principled and impactful.

Our research spans machine learning, computer vision, dynamical systems and robotics, and AI for healthcare. Our machine learning work focuses on deep learning theory, trustworthy AI, deep

generative models, parsimonious representation learning, continual learning, and optimization. In computer vision, we study vision–language models, human motion understanding, video analysis, image analysis, and geometric 3D vision. In dynamical systems and robotics, we develop methods for analyzing and learning linear, hybrid and multi-agent systems. Through close collaborations with clinicians and scientists, we translate advances in AI into systems for radiology, cardiology, neurology, hematology, and surgery. These efforts aim to develop AI technologies that are not only powerful, but also trustworthy and clinically meaningful.

As part of the IDEAS Center, the ASSET Center, and the GRASP Lab at the University of Pennsylvania, we operate at the intersection of theory, engineering, and real-world deployment. We are particularly interested in training students who want to work at the boundary of mathematical foundations and impactful AI systems.

Project Highlights

Many issues: arxiv links wrong, author errors, many preprints are published, etc.

Contact
Prof. René Vidal
University of Pennsylvania
Philadelphia, PA

DATA/CODE

I think something pointing to github repo with data and code would be good. We should somehow port these links and data there. <http://vision.jhu.edu/data/> <http://vision.jhu.edu/code/>

TUTORIALS

We are missing adding the following

<https://achatto1.github.io/Foundations-of-Interpretable-AI/>

<https://iccv.thecvf.com/virtual/2025/tutorial/2853>

<https://cvpr.thecvf.com/virtual/2025/tutorial/35900>

https://2025.cpal.cc/tutorial_info/

PEOPLE

Principal Investigator

René Vidal

Rachleff & PIK University Professor of ESE, Radiology, CIS, Statistics and Data Science
School of Engineering & Applied Science, Perelman School of Medicine, Wharton School
Director of the Center for Innovation in Data Engineering and Science (IDEAS)
Co-Chair of PennAI
University of Pennsylvania

Please modify titles and departmental and school affiliations as per above. Please change my photo to have my face centered and more zoomed in.

Also, I think a webpage for me is missing. Right now my info is in a section called “About”. Please make that a webpage for me. Instead, the content in “About” could be like the content in “Contact us!” <https://vidal-lab.ding.tj/contact/>

Staff and Current PhD Students

I think photos should have faces at the center of the photo and be more zoomed in. I also think many photos have shadows, with the most noticeable being Liangzu, then Fengrui and Uday. There is a typo in the name of Konstantinos Emmanouilidis

Former Research Scientists

Please put start and end dates as well as links to their linkedin so we don’t need to track where they are at.

1. Benjamin Haeffele (2015-2025)
2. Paris Giampouras (2022-2023), now Assistant Professor at University of Warwick
3. Stewart Slocum (2021-2022), now PhD Student at MIT
4. Mahyar Fazlyab (2020-2021), now Assistant Professor at Johns Hopkins University
5. Benjamín Béjar (2017-2019), now Lead Data Scientist, Swiss Data Science Center
6. Haider Ali (2016-2018)
7. Bijan Afsari (2014-2016)

Former Post-doctoral Researchers

Please put start and end dates as well as links to their linkedin so we don't need to track where they are at.

8. Hancheng Min (2023-2025), now Assistant Professor at Shanghai Jiao Tong
9. Paris Giampouras (2022-2023), now Assistant Professor at University of Warwick
10. Mustafa Kaba (2017-2020), then Applied Researcher at eBay, now Applied Scientist at AWS
11. Guilherme Franca (2017-2020)
12. Zhihui Zhu (2018-2019), now Assistant Professor at Ohio State University
13. Manolis Tsakiris (2011-2017), then Assistant Professor at ShanghaiTech, now Chinese Academy of Sciences
14. Shahin Sefati (2014-2016), then Senior Researcher at Comcast, now at Meta
15. Bijan Afsari (2010-2014)
16. Erdem Joruk (2011-2013), now at Vispera Information Technologies Co., Turkey
17. Luca Zapella (2011-2013), then Engineer at Metaio GmbH in Munich, now Engineer Project Manager at Apple in CA, USA
18. Aastha Jain (2011-2012), now at LinkedIn, USA
19. Avinash Ravichandran (2010-2012), now Research Scientist at Amazon
20. Diego Rother (2009-2011), now Engineer at Google, USA
21. Mihaly Petreczky (2006-2007), now Assistant Professor at Ecole des Mines, Douai, France

Former PhD Students

Please put start and end dates as well as links to their linkedin so we don't need to track where they are at.

22. Aditya Cattopadhyay (2018-2024, CS, JHU), now at Applied Scientist at Amazon
23. Ambar Pal (2018-2024, CS, JHU), now at Applied Scientist at Amazon
24. Carolina Pacheco (2018-2024, BME, JHU)
25. Hancheng Min (2019-2023, ECE, JHU, primary advisor Enrique Mallada)
26. Effrosyni Mavroudi (2015-2022, PhD, BME), now at Facebook
27. Tianyu Ding (2017-2021, AMS, JHU, primary advisor Daniel Robinson)
28. Florence Yellin (2015-2020, PhD, ME)
29. Chong You (2012-2018, PhD, ECE, JHU), then postdoc at UC Berkeley, now at Google Research
30. Siddharth Mahendran (2010-2018, PhD, ECE, JHU), now at MagicLeap
31. Evan Schwab (2011-2018, PhD, ECE, JHU, co-advised with Nicolas Charon), then at Philips, now at Johnson and Johnson
32. Giann Gorospe (2009-2017, PhD, BME, JHU)
33. Manolis Tsakiris (2013-2017, PhD, ECE, JHU), now Assistant Professor at ShanghaiTech
34. Colin Lea (2013-2017, PhD, CS, JHU, co-advised with Greg Hager), then at Oculus, now at Apple
35. Lingling Tao (2010-2017, PhD, ECE, JHU), then at Oculus, now at Facebook Reality Labs
36. Ben Haeffele (2013-2015, PhD, BME, JHU, co-advised with Eric Young), now research scientist at JHU
37. Roberto Tron (2007-2012, PhD, ECE, JHU), then post-doctoral researcher at Upenn, now assistant professor at Boston University

38. Rizwan Chaudhry (2006-2012, PhD, CS, JHU), then software engineer at Microsoft, then software engineer at Nest and Google, then manager at Google-Health
39. Ehsan Elhamifar (2006-2012, PhD, ECE, JHU), then post-doctoral researcher at UC Berkeley, then Assistant Professor at Northeastern University
40. Ertan Centigul (2005-2011, PhD, BME, JHU), then research scientist at Siemens Corporate Research, now research director at General Electric
41. Avinash Ravichandran (2004-2010, PhD ECE, JHU), then post-doctoral researcher at UCLA, then research scientist at Amazon
42. Dheeraj Singaraju (2004-2010, PhD ECE, JHU), then post-doctoral researcher at UC Berkeley, then software engineer at Google, now at Waymo
43. Alvina Goh (2004-2010, PhD BME, JHU), then adjunct assistant professor at the National University of Singapore, now Director of AI Platforms at DSAID, GovTech, Singapore.

Former Masters Students

Please put start and end dates as well as links to their linkedin so we don't need to track where they are at.

44. Abhishek Goyal (2024, CIS, Penn) [Linkedin](#)
45. Yutao Tang (2020-2023, BME, JHU)
46. Prashast Bindal (2020-2021, MSc, CS)
47. Connor Lane (2016-2019, MSc, CS)
48. Carolina Pacheco (2016-2018, BME, JHU)
49. Benjamín Béjar (2011-2012, BME, JHU), then post-doctoral fellow at EPFL, then Associate Research Scientist at JHU
50. Jixin Li (2009-2010, MSc, ECE, JHU), now Lead Research Analyst at Videology
51. Gagan Bansal (2007-2008, M.Sc., CS, JHU), then Research Engineering at Yahoo!, then Senior Research Software Development Engineer, now senior RSDE at Azure Machine Learning, Microsoft
52. Atiyeh Ghoreyshi (2005-2006, M.S. BME, JHU), then PhD student at McGill University, then post-doc at University of Southern California, then engineer at Masimo Corporation, then senior R&D scientist at Auris Surgical Robotics

Former Undergraduate Students, Interns / REU Students, Visiting Students

It is hard to distinguish these categories, and my CV vs what you have is not the same. For example, I had intended for undergraduate to mean undergraduates from Penn/JHU who do research while they are students, but issue is that REU students are also undergraduates, just that they come for the summer. Then visiting students can include REUs and interns and PhD students who come to visit. Then most interns did not have a PhD but could have been undergraduates or masters. So, I am thinking, shouldn't we just make a single category? I would perhaps just single out "Visiting Graduate Students" and "High-School students", which are clear distinct categories. Anyhow, here I put what I have in my CV which is to be blended.

Visiting Graduate Students

1. Laura Daza (2022)
2. Julio Hurtado (2020)
3. Salma Tarmoun (2019), now a PhD student at Johns Hopkins University
4. Imke Meyer (2018, Ecole Polytechnique), then PhD student at Ecole Polytechnique, then at INRIA, now at the Institute of Public Health in Berlin
5. Claire Donnat (2015, Ecole Polytechnique), then PhD student at Stanford, now Assistant Professor at the University of Chicago.
6. Islem Rekik (2014)
7. Bertrand Rondepierre (2013, Ecole Normale Supérieure)
8. Laurent Bako (2007), now Associate Professor at Ecole Centrale de Lyon
9. Roberto Tron (2006, Visiting student), then at JHU as a PhD student, then at Penn as Postdoc, now at BU as Assistant Professor

Undergraduate Students

1. Shreya Wadhwa (2022, CS, JHU)
2. Stewart Slocum (2021, CS, JHU)
3. Derek Lim (2020, REU)
4. Alejandro Barrera (2019, REU)
5. Evan Frenklak (2018, REU)
6. Max Lu (2018, REU)
7. Ksenia Lekomtceva (2018, REU)
8. Larissa Clopton (2017, REU)
9. Divya Bhaskara (2016, REU)
10. Alexander Serrano (2015, REU)
11. Quin Liu (2013, REU)
12. Soren Wolfers (2013, DAAD visiting student)
13. Nicolas Jimenez (2012-2013, research assistant)
14. Arunesh Mittal (2012-2013, Neuroscience, JHU)
15. Patrick McClure (2012, REU student)
16. James Breen (2011, REU student)
17. Simon Schütz (2010, DAAD visiting student)
18. Martin Wojkowsky (2010, DAAD visiting student)
19. Aline Elad (2010, REU student)
20. Alex Hsieh (2010, BME, JHU)
21. Lucas Theis (2009, DAAD visiting student)
22. Andreas Beckers (2009, DAAD visiting student)
23. Leyla Isik (2009-2010, REU, BME, JHU), then PhD student at MIT, now Assistant Prof. at JHU.
24. Solomon Liu (2008-2011, BME, JHU)
25. Venkatesh Srinivas (2008, ME, JHU)
26. James Choi (2008, ME, JHU)
27. Charlie Ouyang (2008, B.S. BME, JHU)
28. Vincent Yeh (2007, B.S. BME, JHU)
29. Jai Madhok (2007, B.S. BME, JHU)
30. Matthias Behnisch (2006, Visiting student), now at Bielefeld University, Germany
31. Andy Wong (2006, B.S. BME, JHU)
32. Mary Ellen Pozo (2005, B.S. BME, JHU)

33. Sampreet Niyogi (2004-2005, B.S. BME, JHU), then at the University of Pennsylvania as a PhD student

High School Students

34. Iraj Shroff (2025)

35. Prisha Shroff (2024)

36. Sruti Nuthalapati (2010)

37. Maddie Crowl (2009), Women in Science and Engineering (WISE) program

38. Andy Tien (2006), then at JHU as an undergraduate student

Teaching

Research

Pls add links to Fall 2024 ESE-6450-001 Deep Generative Models

Machine Learning

We develop the theoretical and algorithmic foundations of modern machine learning systems.

Deep learning theory

We study the mathematical principles behind modern neural networks. Our lab focuses on understanding the optimization landscape and generalization properties of positively homogeneous networks, analyzing the learning dynamics and implicit bias of gradient-based methods, and providing principled explanations to deep learning phenomena such as neural collapse, low-rank adaptation, and learning on the edge of stability. Through this work, we aim to connect fundamental theory with the practical behavior of large-scale neural networks.

Trustworthy AI

We develop AI systems that are interpretable, robust, and reliable for high-stakes applications. Our information pursuit framework enables explainable-by-design models that make predictions from a sequence of informative question–answer pairs. Our parsimonious concept engineering method improves the safety of multimodal generative models through targeted interventions in their latent spaces. Our theoretical analysis of the conditions under which robust classifiers are guaranteed to exist enables the design of provably robust classifiers. Our algorithms for generating adversarial attacks stress test the reliability of visual classifiers and large language models. This research lays the foundation for trustworthy AI in high-stakes applications.

Deep generative models

We explore the theory and practice of deep generative models. Our contributions include convergence guarantees for generative model inversion and transformer-based diffusion models. Our work on flow matching aligns latent distributions with the multimodal structure of

real data, improving optimization and convergence. We have also developed methods for accurate diffusion-based image restoration and for fast, high-quality text-to-image generation via retrieval-augmented generation with few-step diffusion models. These efforts provide principled methods for efficient and reliable generative modeling.

Parsimonious representation learning

We focus on discovering sparse and low-rank structure in high-dimensional data. Our lab pioneered the development of methods for clustering data in a union of subspaces, including methods such as Generalized Principal Component Analysis (GPCA), Sparse Subspace Clustering (SSC), and Dual Principal Component Pursuit (DPCP). We also develop nonconvex matrix and tensor factorization methods with guarantees of global optimality. This work provides scalable, robust, and geometrically principled tools for representation learning.

Continual learning

We build models that learn tasks sequentially without forgetting. By linking continual learning with adaptive filtering, we leverage classical theory to inform new algorithms. Frameworks like the Ideal Continual Learner (ICL) and LoRanPAC unify existing methods, provide theoretical guarantees, and maintain stability over long task sequences. Our goal is to develop continual learning methods that are both theoretically sound and practically effective.

Optimization

We analyze the convergence and global optimality of optimization algorithms for machine learning, including problems such as subspace clustering, nonconvex matrix and tensor factorization, dictionary learning, generative model inversion, and deep learning. We also study distributed optimization on Riemannian manifolds and accelerated methods for both smooth and nonsmooth problems from a dynamical systems perspective. Through this work, we unify theory and practice to deliver principled, scalable optimization algorithms for machine learning.

Computer Vision

We develop methods for visual perception that combine geometry, learning, and multimodal reasoning.

Vision–Language Models

We develop vision–language models that connect visual data with semantic concepts to enable interpretable and controllable multimodal AI systems. Our work includes methods for visual grounding, image captioning, and multimodal generation that explicitly model the relationship between images, language, and structured knowledge. We have developed concept-based interpretable classification frameworks that explain model predictions through human-understandable visual attributes. We have also introduced approaches such as knowledge pursuit prompting for zero-shot multimodal synthesis and concept-based image editing methods that enable targeted manipulation of visual content. Our research further explores hierarchical concept representations to improve interpretability and robustness. This work advances multimodal AI systems that are interpretable, controllable, and robust.

Human Motion Analysis

We develop methods for understanding human motion from video and skeletal data. Our work spans action recognition, detection, and segmentation, as well as human pose estimation, combining geometric modeling, structured temporal representations, and deep learning. Early contributions to action recognition combined kernels on dynamical systems with optical-flow-based feature representations. Subsequent work established benchmarks such as the MHAD dataset and introduced methods based on informative skeletal joints and moving poselets. Later work developed spatio-temporal convolutional networks and temporal convolutional models for action detection and segmentation, as well as visual-symbolic representations for recognizing complex group activities. More recently, we have developed methods for human pose estimation, disentangled motion representation learning, and multimodal motion representations that support recognition, retrieval, and generation of human movements. This work provides principled and data-driven approaches for fine-grained human behavior understanding.

Video Analysis

We study methods for modeling dynamic visual phenomena in videos. Our work includes early contributions to modeling dynamic textures through dynamical systems formulations, enabling registration, recognition, and tracking of complex temporal patterns. We have also developed efficient approaches for semantic video segmentation that capture the spatiotemporal structure of visual processes. More recent work explores the robustness of video classification systems through adversarial training, and the generation of perpetual videos of dynamic scenes while enforcing 3D consistency. By combining geometric representations, dynamical systems, and deep learning, this research provides principled tools for understanding complex temporal patterns in visual data.

Image Analysis

We develop algorithms for image segmentation, restoration, and visual representation learning. Our work includes foundational contributions to image segmentation and alpha matting based on discrete optimization, semantic segmentation methods that combine structured output learning and conditional random fields, and visual representation learning methods that improve image categorization and clustering. We have also developed techniques for image deblurring and restoration based on both classical image priors and modern priors derived from diffusion models. This research provides robust and scalable tools for extracting meaningful structure from visual data and improving image quality for downstream tasks.

Geometric Vision

We develop geometric methods for recovering 3D structure and motion from images and video. Our early work introduced algebraic and geometric approaches for multibody motion segmentation, including formulations based on multibody epipolar geometry and higher-order tensor representations. This work also led to widely used benchmarks such as the Hopkins 155 dataset. Subsequent contributions developed robust methods for motion segmentation, 3D registration, and object pose estimation, as well as optimization-based approaches for geometric estimation using tools such as dual principal component pursuit and iteratively reweighted least squares. Our recent work continues to advance robust geometric estimation

for rotation search and 3D reconstruction. This research establishes mathematically grounded approaches to geometric vision with strong robustness and optimality properties.

Dynamical Systems and Robotics

We develop mathematical tools for modeling, optimization, and control of complex dynamical systems.

Learning and Analyzing Linear, Bilinear and Hybrid Dynamical Systems

We develop rigorous methods for analyzing and learning dynamical systems. Our work on hybrid systems established conditions for observability and algorithms for system identification, providing rigorous tools for modeling systems that exhibit both continuous and discrete behaviors. We have also developed realization theory for stochastic jump-Markov linear systems and stochastic bilinear systems, providing fundamental characterizations of when such systems admit minimal and identifiable representations. More recent work studies observability and identification of linear systems with sparse inputs, developing methods to recover system dynamics when inputs are high-dimensional but only sparsely active. These contributions provide new tools for identifying dynamical systems from limited data.

Geometry and Distances of Spaces of Dynamical Systems

We develop principled distances and similarity measures for comparing dynamical systems. Our early work introduced Binet–Cauchy kernels for comparing linear dynamical systems, reducing kernels on trajectories to kernels on the parameters. Building on this foundation, we developed group-action-induced distances for averaging and clustering dynamical systems, establishing a geometric framework that accounts for invariances in system representations. This work was further extended through the development of alignment distances on spaces of dynamical systems, together with efficient algorithms for their computation. These contributions have been particularly impactful in computer vision, where dynamical systems provide compact representations of temporal data, enabling robust methods for video comparison, dynamic scene understanding, and human action recognition.

Dynamical Systems Perspectives on Optimization Algorithms

We investigate optimization algorithms through the lens of dynamical systems. By modeling classical and accelerated gradient-based methods as continuous-time dynamical systems, we obtain new insights into convergence, stability, and robustness. This includes studies of accelerated gradient descent, ADMM, conformal symplectic and relativistic optimization, nonsmooth dynamical systems, and proximal splitting methods. Our approach unifies theory and practice, providing both principled explanations of algorithmic behavior and design of novel, efficient optimization methods.

Distributed Optimization and Consensus on Manifolds

We develop geometric methods for distributed optimization and consensus on manifolds. Our early work introduced Riemannian optimization methods for optimal motion estimation on the

essential manifold, establishing geometric formulations for recovering camera motion from image correspondences. Building on this foundation, we developed convergence guarantees for gradient descent methods for computing Riemannian centers of mass, providing fundamental tools for averaging data on manifolds. This work was further extended through the development of distributed consensus algorithms and consensus on manifolds, together with theoretical guarantees and scalable algorithms for multi-agent coordination. These contributions have been particularly impactful in computer vision, robotics, and biomedical imaging. In vision and robotics, they have enabled distributed localization of camera sensor networks, coordinated estimation, and geometric averaging methods for multi-view perception and autonomous systems. In brain imaging, these methods have also enabled principled averaging and clustering of probability density functions arising in high angular resolution diffusion MRI, supporting applications such as population analysis and morphometry.

Robotics and Autonomous Systems

We develop geometric and control-theoretic methods for autonomous robotic systems operating in complex environments. Our early work introduced vision-based control strategies for autonomous helicopter landing, demonstrating how visual feedback can be integrated with dynamical models to enable precise navigation and control. We also developed game-theoretic formulations for pursuit–evasion games, providing strategies for multi-agent coordination in adversarial settings. These contributions established principled connections between vision, control, and multi-agent systems, and helped advance the development of autonomous robots capable of perception-driven decision making and coordinated behavior.

AI for Healthcare

We translate advances in AI into clinically meaningful tools for diagnosis, monitoring, and biomedical discovery.

Diffusion MRI Analysis

We develop computational methods for the analysis of high angular resolution diffusion imaging (HARDI) data to study brain structure and connectivity. Our work includes foundational contributions to HARDI reconstruction, restoration, registration, and segmentation based on information geometry, sparse representation, and non-convex optimization. A key contribution of our work is the ability to process diffusion signals while respecting the underlying geometric structure of the space of orientation distribution functions. These methods have enabled applications such as the study of structural brain asymmetries in twins and the identification of imaging biomarkers for neurological disorders such as mild cognitive impairment. This work provides principled tools for extracting clinically meaningful information from high-dimensional neuroimaging data.

Surgical Activity Recognition and Skill Assessment

We develop computer vision and machine learning methods for understanding surgical workflows and assessing surgical skill from robotic surgery data. Our work includes early

contributions based on bag-of-spatiotemporal feature representations, sparse hidden Markov models, and conditional random fields for gesture segmentation and classification. We have also developed spatio-temporal deep learning architectures for surgical activity recognition and skill assessment. In addition, our lab contributed to the development of widely used benchmarks, such as the JHU-ISI gesture and skill assessment dataset, and helped shape the field through influential tutorials on data-driven methods for robotic surgery. This research provides quantitative and automated tools for objective surgical training and assessment.

Analysis of Stem Cell–Derived Cardiomyocytes

We develop machine learning methods for the analysis of stem cell–derived cardiomyocytes to support advances in regenerative medicine and drug discovery. Our work includes approaches for clustering and classification of cardiac cells based on cell morphology and contractile dynamics. We have developed shape analysis methods based on metamorphosis models to capture complex cell deformations, as well as recurrent neural network models to analyze temporal contraction patterns. These methods enable automated phenotyping of cardiac cells and provide scalable tools for understanding variability in stem-cell-derived cardiac populations.

Computational Microscopy and Point-of-Care Diagnostics

We develop computational imaging and machine learning methods for cell detection, classification, and counting from holographic microscopy data. Our work includes sparse representation-based approaches for holographic reconstruction, hybrid physics-based and deep learning methods for phase retrieval, and methods that combine dictionary learning with graphical models for cell detection, classification, and counting. More recent work has explored encoder–decoder deep learning architectures for robust detection and counting of cells in holographic images. A major focus of this research is the development of computational tools for point-of-care diagnostics, including automated complete blood count with three-part differential and urinalysis using lensless imaging platforms. This work advances low-cost and scalable diagnostic technologies for resource-constrained healthcare settings.

Computer Vision for Neurological and Developmental Disorders

We develop computer vision methods for the analysis of human movement to support the diagnosis and treatment of neurological and developmental disorders. Our work includes methods for recognizing infant actions from multi-view video to support rehabilitation therapy. We have also developed deep learning models for detection and segmentation of motor tics from video for Tourette syndrome assessment. A major focus of our recent work is the Computerized Assessment of Motion Imitation (CAMI), which uses both 3D skeletal motion capture data and 2D video-based pose estimation to quantify motor imitation ability as a biomarker for autism spectrum disorder. These methods enable objective and quantitative assessment of motor function and have demonstrated promise for distinguishing typically developing children from children with autism. This research establishes computer vision as a tool for scalable and objective behavioral assessment in clinical settings.

Medical Vision–Language Models and Foundation Models for Healthcare

We develop multimodal AI systems that combine medical images and clinical text to enable interpretable and reliable clinical decision support. Our work includes methods for extracting

structured medical facts from radiology reports through learned text representations and using these representations to build interpretable-by-design systems for report classification. Our recent work extends these ideas to jointly reason over medical images and reports by grounding extracted clinical concepts directly in chest X-ray images, enabling automated report generation and interpretable classification. More broadly, we are extending these methods to other imaging modalities such as echocardiography, with the goal of developing foundation models for radiology and cardiology. This research advances multimodal foundation models for healthcare that emphasize interpretability, reliability, and clinical utility.

PROMPT FOR CREATING RESEARCH BLURBS IN MACHINE LEARNING

I will give you three paragraphs. I want you to write a paragraph in the style and length of paragraphs (1)-(3), but with the specific details and flow from paragraph (3). Here are the paragraphs.

I will give you four paragraphs. I want you to write a paragraph in the style of paragraphs (1)-(3) but shorter, with the background knowledge of paragraph (4), but with the specific details and flow from paragraph (5). Here are the paragraphs.

- (1) Deep learning theory encompasses the foundational mathematical principles that underpin modern neural networks and their capabilities. Our lab has made foundational contributions to understanding the optimization landscape of positively homogeneous networks, the learning dynamics and implicit bias of gradient-based methods, and the generalization properties of deep networks. Our current efforts are directed toward understanding the dynamics of low-rank adaptation methods and training at the edge of stability. Through this work, we aim to connect fundamental theory with the practical behavior of large-scale neural networks, advancing principled approaches to the design and analysis of modern AI systems.
- (2) Trustworthy machine learning focuses on developing reliable and accountable AI systems for high-stakes applications. Our lab has made fundamental contributions to interpretability and robustness of AI. Our information pursuit framework enables explainable-by-design models that make predictions through sequences of the most informative question-answer pairs. Our work on parsimonious concept engineering improves the safety of generative models through targeted interventions in the latent representations of foundation models. Our work on robust classification establishes conditions on the data under which robust classifiers are guaranteed to exist, enabling the design of provably robust classifiers. Through this work, we develop theoretical and algorithmic foundations for interpretable, robust, and trustworthy AI systems.
- (3) Deep generative models aim to learn the structure of complex data distributions for high-fidelity data generation, yet their theoretical foundations remain limited. Our lab has made fundamental contributions to the theory of generative modeling, including convergence guarantees for generative model inversion and transformer-based diffusion models. In addition, our work on flow matching improves optimization and convergence rates by aligning latent distributions with the multimodal structure of real data. Through this work, we develop principled theoretical foundations for more efficient and reliable generative modeling algorithms.
- (4) Parsimonious Representation Learning focuses on discovering compact, efficient ways to represent complex data while preserving essential information. Matrix factorization techniques decompose high-dimensional data matrices into lower-dimensional components, revealing latent structures and enabling applications like recommendation systems and dimensionality reduction. Subspace clustering methods identify and group data points that lie near lower-dimensional linear or affine subspaces within the ambient space, allowing for more accurate clustering of high-dimensional data with complex geometric structures. Manifold learning approaches discover nonlinear, low-dimensional structures that capture the intrinsic geometry of data, assuming that high-dimensional observations often lie on or near a lower-dimensional manifold, thus enabling more effective visualization, compression, and feature extraction while respecting the underlying data geometry.

- (5) You should mention Rene Vidal's seminal work on Generalized Principal Component Analysis, continuing with Ehsan Elhamifar work on Sparse Subspace Clustering, which was extended by Chong You to allow for scalability and robustness. Then, there is the great work of Haeffele on matrix factorization via non-convex formulations with variational norm regularization, followed by the work Manolis Tsakiris, Zhihui Zhu and Tianjiao Ding on Dual Principal Component Pursuit. Please search the literature on this to make a nice really great integrated paragraph.

PROMPT FOR CREATING RESEARCH BLURBS IN COMPUTER VISION

I am creating a webpage for the research group of Professor Rene Vidal and I want you to write multiple paragraphs for his research in computer vision. I will give you (1) one paragraph summarizing all the research areas in Vidal Lab. (2) Five paragraphs describing sub-areas of research in machine learning conducted by the Vidal Lab, which you should use as examples. (3) Multiple paragraphs with a list of papers in sub-areas of computer vision that the Vidal Lab works on. Using the machine learning examples in (2), please write a paragraph for each group of papers in (3). Here are the paragraphs.

(1)

The Vidal Lab conducts research on machine learning, computer vision, dynamical systems, robotics, and biomedical data science. Our machine learning research focuses on deep learning theory, trustworthy AI, deep generative models, parsimonious representation learning, continual learning, and optimization. In computer vision, we focus on vision-language models, 3D vision, video analysis, and human motion analysis. In robotics and dynamical systems, we focus on linear, hybrid and multi-agent systems. These techniques drive medical applications in radiology, cardiology, hematology, and surgery. As part of the Center for Innovation in Data Engineering and Science (IDEAS), the Center for AI-Enabled Systems: Safe, Explainable & Trustworthy (ASSET), and the General Robotics, Automation and Perception Lab (GRASP), we integrate theoretical breakthroughs with practical, impactful solutions.

(2)

Deep learning theory

We study the mathematical principles behind modern neural networks. Our lab focuses on understanding the optimization landscape and generalization properties of positively homogeneous networks, analyzing the learning dynamics and implicit bias of gradient-based methods, and providing principled explanations to deep learning phenomena such as neural collapse, low-rank adaptation, and learning on the edge of stability. Through this work, we aim to connect fundamental theory with the practical behavior of large-scale neural networks.

Trustworthy AI

We develop AI systems that are interpretable, robust and reliable for high-stakes applications. Our information pursuit framework enables explainable-by-design models that make predictions from a sequence of informative question–answer pairs. Our parsimonious concept engineering method improves the safety of multimodal generative models through targeted interventions in their latent spaces. Our theoretical analysis of the conditions under which robust classifiers are guaranteed to exist enables the design of provably robust classifiers. Our algorithms for generating adversarial attacks stress test the reliability of visual classifiers and large language models. This research lays the foundation for trustworthy AI in high-stakes applications.

Deep generative models

We explore the theory and practice of deep generative models. Our contributions include convergence guarantees for generative model inversion and transformer-based diffusion models. Our work on flow matching aligns latent distributions with the multimodal structure of real data, improving optimization and convergence. We have also developed methods for accurate diffusion-based image restoration and for fast, high-quality text-to-image generation

via retrieval-augmented generation with few-step diffusion models. These efforts provide principled methods for efficient and reliable generative modeling.

Parsimonious representation learning

We focus on discovering sparse and low-rank structure in high-dimensional data. Our lab pioneered the development of methods for clustering data in a union of subspaces, including methods such as Generalized Principal Component Analysis (GPCA), Sparse Subspace Clustering (SSC), and Dual Principal Component Pursuit (DPCP). We also develop nonconvex matrix and tensor factorization methods with guarantees of global optimality. This work provides scalable, robust, and geometrically principled tools for representation learning.

Continual learning

We build models that learn tasks sequentially without forgetting. By linking continual learning with adaptive filtering, we leverage classical theory to inform new algorithms. Frameworks like the Ideal Continual Learner (ICL) and LoRanPAC unify existing methods, provide theoretical guarantees, and maintain stability over long task sequences. Our goal is to develop continual learning methods that are both theoretically sound and practically effective.

Optimization

We analyze the convergence and global optimality of optimization algorithms for machine learning, including problems such as subspace clustering, nonconvex matrix and tensor factorization, dictionary learning, generative model inversion, and deep learning. We also study distributed optimization on Riemannian spaces, and accelerated methods for both smooth and nonsmooth problems from a dynamical systems perspective. Through this work, we unify theory and practice to deliver principled, scalable optimization algorithms for machine learning.

(3)

Human Motion Analysis

Work of Rizwan Chaudhry on action classification (CVPR 2012), sequence of most informative joints (JVCIR 2014), MHAD database (WACV 2013), action recognition (CVPR 2009, CVPR 2013, CDC 2012, CDC 2013). Work of Lingling Tao on temporal video segmentation (IPCAI 2012, MICCAI 2013), moving poselets (CVPR 2015). Work of Colin Lea on action segmentation and detection with spatio-temporal CNNs, Temporal Convolutional Networks and Segmental Spatio-Temporal CNNs. (CVPR 2017, ECCVW 2016, ECCV 2016, ICRA 2016, WACV 2015). Work of Efi Mavroudi on moving poselets (WACV 2017), action detection (WACV 2022), visual-symbolic graphs for recognizing groups of actions (ECCV 2020), semantic segmentation of fine-grained actions (2018). Work of Kaleab Kinfu on human pose estimation (BMVC 2023), Assessment of motor imitation (TBME 2025), MotionBind for retrieval, recognition and generation (NeurIPS 2025).

Vision Language

Work of Efi Mavroudi on attend, tell and ground (2021) and visual grounding and image captioning (CVPR 2022). Work of Jinqi Luo on Knowledge Pursuit Prompting for Zero-Shot Multimodal Synthesis (ECCVW 2024) and image editing via Concept Lancet (CVPR 2025). Work of Nghia Nguyen on Sparse Semantic Concept Defenses Against Adversarial Attacks (ICLRW 2026) and Hierarchical concept embeddings of (CVPR 2026). Work of Aditya Chattopadhyay on

interpretable classification via learning concepts (NeurIPS 2023, ICLR 2024) in collaboration with Armand Comas (ICLR 2025) and Stefan Kolek (CVPR 2025).

Video Analysis

Work of Avinash Ravichandran on registration of dynamic textures (CVPR 2005, ECCV 2008), recognition of dynamic textures (CVPR 2009, ACCV 2010, TPAMI 2012). Work of Rizwan Chaudhry on dynamic template tracking (IJCV 2013). Work of Aastha Jain on semantic video segmentation (ICCV 2013). Kaleab Kinfu on adversarial training for video (CVPR 2022). Work of Fengrui Tian on Voyaging into Perpetual Dynamic Scenes from a Single View (ICCV 2025).

Image Analysis

Work of Deeraj Singaraju on alpha matting (TPAMI 2010), image segmentation (CVPR 2008, CVPR 2009), semantic segmentation (CVPR 2011). Work of Avinash Ravichandran on Image Deblurring (BMVC 2012). Work of Lingling Tao on semantic segmentation (ECCV 2013). Work of Hans Lobel on visual representation learning for image categorization (CVPR 2013, TPAMI 2015, CVIU 2020). Work of Tianjiao Ding on image clustering (Work of Darshan Thaker on image restoration (CVPR 2025).

Geometric Vision

Vidal's work on optimal motion estimation (ICCV 2001, CIS 2001), motion segmentation (CVPR 2003, CVPR 2004, ECCV 2004, ICRA 2005, CVPR 2005, JMIV 2005, ACCV 2006), including work with Hartley on multibody fundamental matrices and multibody trifocal tensor. Work of Roberto Tron on 3D motion segmentation (CVPR 2008, IJCV 2008), including the Hopkins 155 database (CVPR 2007), work with Rao Tron and Ma on robust motion segmentation (CVPR 2008, TPAMI 2009). Work of Deeraj Singaraju on motion segmentation (CVPR 2005, ACCV 2006, CVPR 2007). Work on object pose estimation by Siddharth Mahendran (ICCV 2017, ECCVW 2018). Work of Tianjiao Ding on applying DPCP to geometric vision (CVPR 2020), Liangzu Peng on IRLS (CVPR 2023), ARCS (CVPR 2022), 3D registration (CVPR 2024), robust rotation search (ECCV 2022).

PROMPTS FOR CREATING RESEARCH PAGE ON DYNAMICAL SYSTEMS AND CONTROLS

I need you to write a few paragraphs for the research webpage about the work of the Vidal Lab on dynamical systems, robotics, optimization on manifolds for distributed consensus problems and dynamical systems approaches for analyzing optimization algorithms. I am struggling on how to organize it, so please think carefully about it. One option is to group them as (1) All the work on observability, reachability identification, realization of both linear systems with sparse inputs, bilinear systems, jump-Markov systems, and hybrid systems, (2) all of the work in robotics, including vision-based landing, pursuit evasion games, formation control, and camera networks, (3) all of the work on optimization on manifolds, consensus, (4) all of the work on dynamical systems for optimization. But I am not sure if that is the best, since some camera localization work can suit robotics as well, although that work is also related to consensus on manifolds. Here is the list

Rene Vidal's work on hybrid systems observability, reachability, identification (HSCC 2000, CDC 2001, CDC 2002, CDC 2003, CDC 2004, HSCC 2003, ACC 2004, HSCC 2005, ECC 2007, EJC 2007, HSCC 2008, Automatica 2008, ACC 2009, CDC 2010, Automatica 2011, HSCC 2011). Work of Petreczky on realization theory for stochastic jump-Markov linear systems and stochastic bilinear systems (CDC 2007, HSCC 2008, TAC 2017). Work on observability and identification of linear systems with sparse inputs of Shahin Sefati (ACC 2015) and Kyle Poe (CDC 2022, CDC 2023). Here are some details for a subset of the work <http://www.vision.jhu.edu/hybrid.htm>

Rene Vidal's work on vision-based landing of a helicopter (ICRA 2001, ICRA 2002), pursuit evasion games (ICRA 2001, CDC 2001, TRA 2002), formation control (ICRA 2003, IROS 2003).

Geometry and Distances of Spaces of Dynamical Systems: I want the center of gravity of the paragraph to be the development of distances for comparing dynamical systems, starting with the work of Vidal on Binet-Cauchy kernels (IJCV 2006), continuing with the group action induced distances, and ending with alignment distances. Specifically, please highlight Bijan Afsari's work on group action induced distances for averaging and clustering dynamical systems (CVPR 2012), alignment distance on spaces of dynamical systems (CDC 2013), algorithms for computing such distances (ECC 2013), and the survey (geometric theory of information 2014). I also want you to emphasize how this work has been extremely impactful in computer vision applications, specifically for comparing videos and human action recognition. Here are some more details: <http://www.vision.jhu.edu/distancesDynamicalSystems.htm>

Following the same style, I want you to write a paragraph about Distributed Optimization and Consensus on Manifolds. The content should include René Vidal's work on optimal motion estimation via Riemannian optimization in the essential manifold (ICCV 2001), Bijan Afsari's work on convergence of gradient descent for finding riemannian center of mass (SIAM-JCO 2013), and Roberto Tron's work on distributed consensus (CDC 2010, ACC 2011) and consensus on manifolds (CDC 2011, CDC 2012, TAC 2012, SIAM JCO 2013). Here are some more details for a subset of the work <http://www.vision.jhu.edu/consensus.htm>. I think highlighting that this work has had impact on both vision and robotics, specifically Roberto Tron's work on distributed localization of camera sensor networks (CDSC 2008, CDC 2009, TAC 2014), would be great.

Guilherme Franca (ICML 2018, Journal of statistical mechanics 2020 and 2021, TAC 2023, Physical Review 2021) on a dynamical systems perspective on various accelerated gradient descent methods, including ADMM, conformal symplectic and relativistic optimization,

nonsmooth dynamical systems, dissipative symplectic integrators, gradient flows and proximal splitting methods.

PROMPTS FOR CREATING RESEARCH PAGE IN BIOMEDICAL DATA SCIENCE

I need you to write a few paragraphs for the research webpage about the work of the Vidal Lab on biomedical data science, including work on diffusion MRI, gesture and skill recognition in robotic surgery, classification of stem cell derived cardiac myocytes, cell detection, classification and counting in holographic images, computer vision for pediatric rehabilitation, computer vision for Tourette syndrome diagnosis, computer vision for autism diagnosis. Here are the details.

First the work of Alvina Goh (Neuroimage 2011, MICCAI 2009, ECCV 2008), Ertan Cetinul (TMI 2013, ISBI 2012) and Evan Schwab (IPMI 2013, MICCAI 2016, SIAM-IS 2019, MedIA 2018, MICCAI 2012, ISBI 2014, MICCAI 2018), on analysis of high angular resolution diffusion images (HARDI), mentioning both the problems, like HARDI reconstruction, processing, registration, and segmentation, as well as the applications, like the twin asymmetry study or the work on mild cognitive impairment. Old text for background: <http://www.vision.jhu.edu/hardi.htm>

Second, the work of Benjamin Bejar Haro and Luca Zapella (MedIA 2013, MICCAI 2012 best paper award), Ehsan Elhamifar (IPCAI 2012), Lingling Tao (MICCAI 2013), Colin Lea (WACV 2015) on bag of spatio-temporal features, sparse hmms, conditional random fields, and spatio-temporal networks for segmentation and recognition of surgical gestures and assessment of surgical skill. Please also highlight the JHU-ISI dataset (M2CAI 2014) and the tutorial on robotics surgery by Narges Ahmidi (TBME 2017).

Third, the work of Giann Gorospe (Circulation 2012, MICCAI 2013, TBME 2014, MICCAI 2015) and Carolina Pacheco (MICCAI 2019) on classification and clustering of stem cell-derived cardiomyocytes using shape analysis (metamorphosis) or recurrent neural networks.

Fourth, the work of Benjamin Haeffele (ISBI 2017, MICCAI 2017), Florence Yellin (ISBI 2017, CVPR 2018, Workshop on Machine Learning in Medical Imaging 2017), Carolina Pacheco (TBME 2022, Optic Express 2022, MICCAI 2024) on sparse representation based reconstruction of holographic images, integration of physics based and deep learning approaches for holographic reconstruction and PSD estimation, integration of dictionary learning and graphical models for cell detection and counting, as well as encoder-decoder architectures for cell detection and counting in holographic images, and maybe most notably the application to point of care testing and diagnostics for complete blood count with three part differential as well as urinalysis. Old text for background: <http://vision.jhu.edu/lfi.htm>

Fifth, the work of Effrosyni Mavroudi (Journal of Neuroengineering and Rehabilitation 2020) and Carolina Pacheco (ICPR 2021) on infant action recognition from multiple videos for rehabilitation therapy. Then maybe the work of Yutao Tang (ICPR 2022) on tic detection for Tourette syndrome diagnosis using TicNet for action detection and segmentation. Then, and this is probably the main work to be highlighted out of the three, the Computerized Assessment of Motion Imitation (CAMI-3D) work of Carolina Pacheco (Biological Psychiatry 2021, Autism Spectrum Disorders 2021, Imaging Neuroscience 2025, British Journal of Psychiatry 2025) and the CAMI-2D net extension of Kaleab Kinfu (to be published in TBME 2026), which use both 3D skeletal data and video data, respectively, to assess motor imitation performance and use that as a biomarker for distinguishing normally developing from autistic children.

Finally, the newest work from Pablo Messina (ACL 2024) on extracting and encoding medical facts from chest radiology reports via learning radiological text representations, and Yuyan Ge (MICCAI 2025) on using such facts to build an interpretable by design system for classification of chest radiology reports. We are in the process of extending both works to work from chest-xray images, to both ground the facts in the images, automatically generate reports from images, and do interpretable by design classification using information pursuit ideas. More broadly, we are also moving to doing similar tasks from ecocardiograms, so the lab is moving towards foundation models in radiology and cardiology.